



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SELVAM FUELS IOC DEALERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

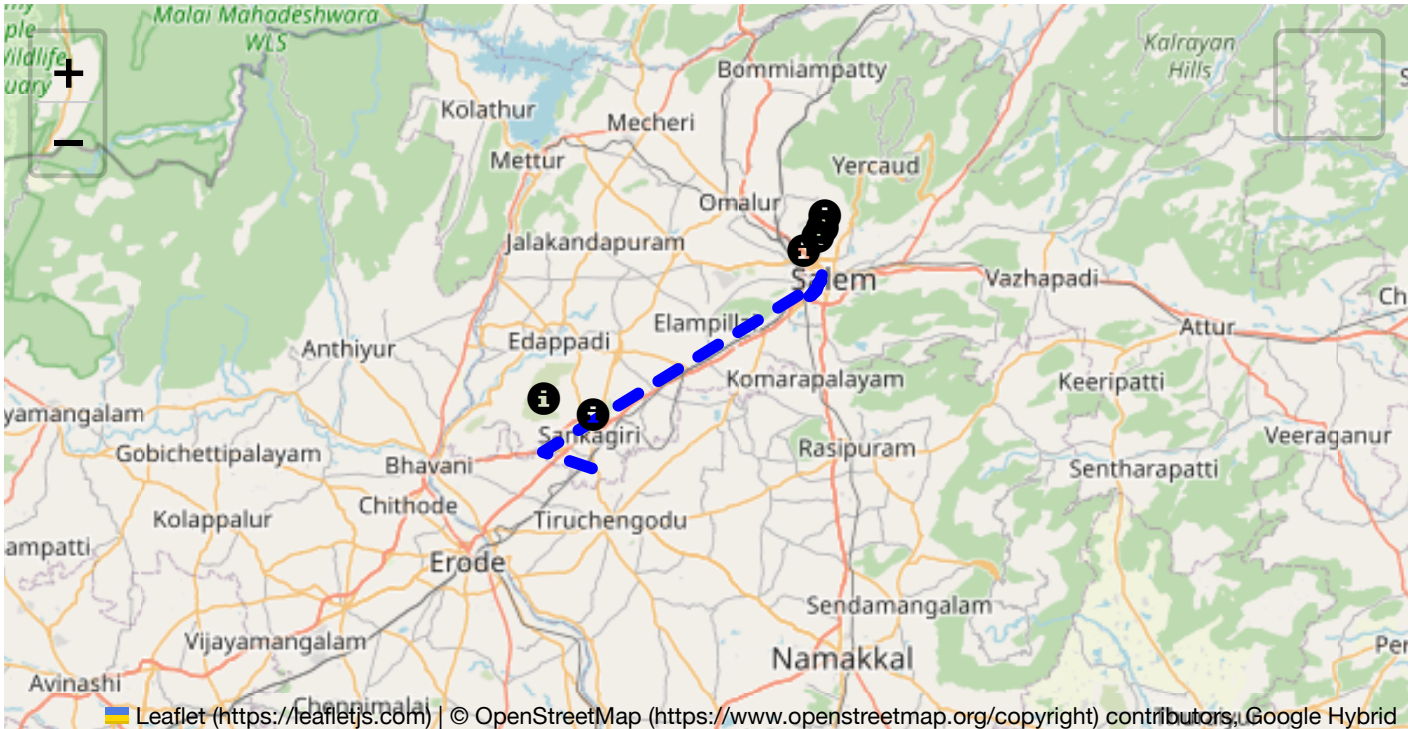
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

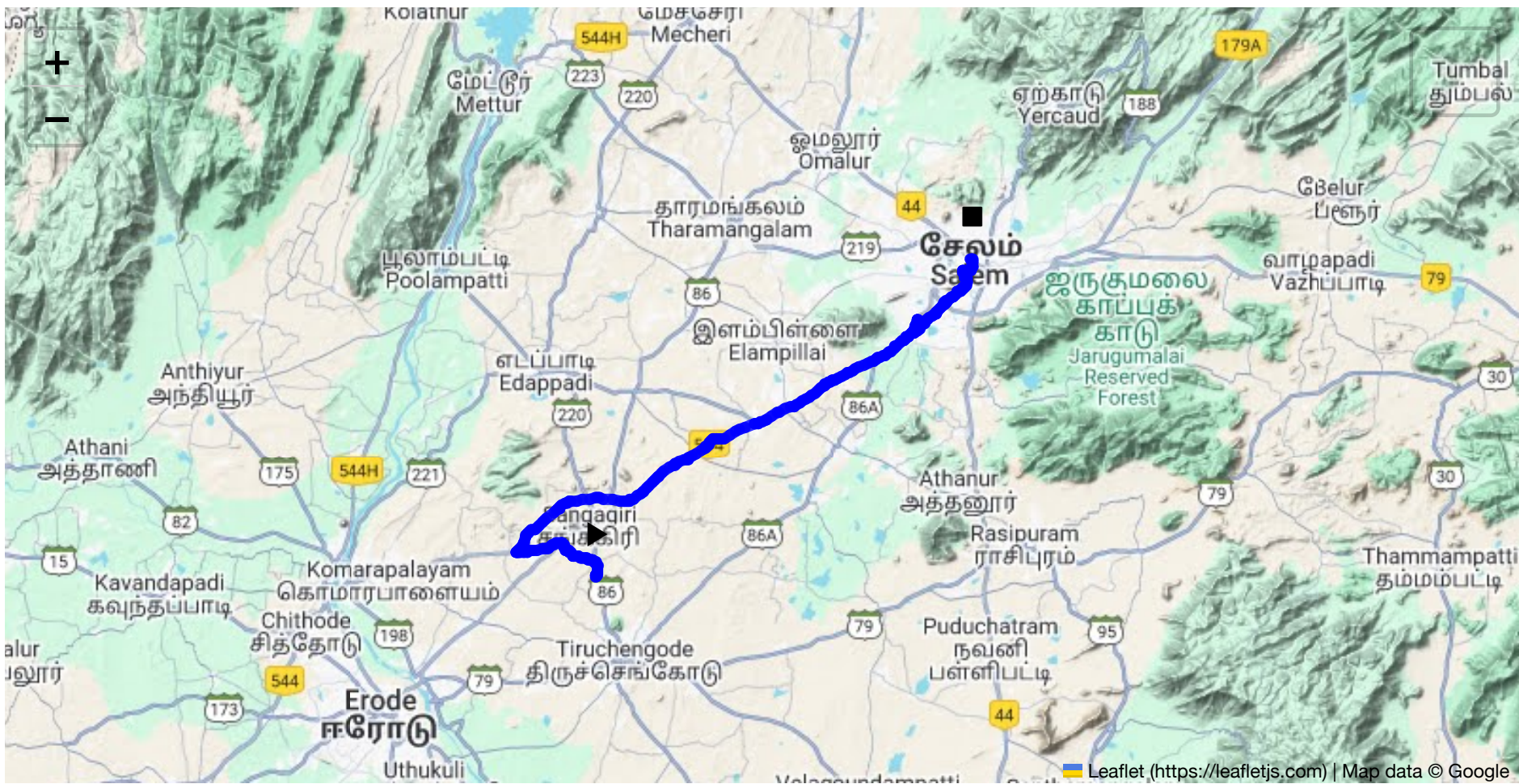
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 56.52 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours

Start: (11.4381, 77.8734)
End: (11.664597, 78.145807)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to M47W+V79, Parvathipuram in Salem covers approximately 56.52 kilometers. This journey involves traversing through small towns and urban areas, notably via Bhavani Main Road, Sandaipet Main Road, and Sankari Main Road. The route primarily runs through rural and semi-urban settings, transitioning into urban areas as it approaches Salem.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The weather in this region is typically tropical, with a hot and dry climate for most of the year. During the monsoon months (October to December), heavy rainfall could cause waterlogging and slippery roads, particularly in poorly drained areas. Fog can also be an issue early in the morning during the winter months, potentially reducing visibility.

3. Analysis of Traffic Patterns

The route experiences variable traffic conditions. Heavy traffic is common during morning (7:30 AM - 9:30 AM) and evening rush hours (5:30 PM - 7:30 PM), especially as you approach urban areas like Salem. Congestion-prone areas include market streets and junctions in Shevapet and Kondalampatti due to local business activities and narrow roads.

4. Assessment of Road Quality and Infrastructure

Road quality varies along the route. While some sections are well-maintained, others, particularly in rural areas, might suffer from potholes and narrow lanes. Urban segments in Salem might have better infrastructure but are subject to traffic congestion potentially affecting travel speed and safety.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, diversions could include:

- Utilizing state or national highways parallel to the route, such as NH544 or NH44, offering better road conditions and faster travel times.
- Local road networks that bypass major congestion zones, potentially reducing delay.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials through Tamil Nadu requires adherence to the CMV Rules and local safety regulations enforced by state authorities. Restrictions on movement during high-peak traffic hours in urban areas might apply, particularly in Salem.

7. Overview of Historical Incidents

There have been occasional reports of incidents involving heavy vehicles on roads near Salem, mainly due to mechanical failures or driver fatigue. Some may involve hazardous materials, emphasizing the need for strict compliance with safety norms.

8. Environmental Considerations and Sensitive Areas

The route passes at proximity to agricultural lands and urban residential areas, necessitating careful handling of hazardous materials to prevent accidental spills. Noise pollution in urban centers could also be a concern.

9. Analysis of Communication Coverage

Overall, mobile network coverage is generally robust along this route, but potential dead zones might exist in rural stretches. Ensuring reliable communication means carrying alternative devices or emergency communication tools.

10. Estimated Emergency Response Times

Emergency response times vary:

- Urban area response, particularly in Salem: Approximately 15-30 minutes.
- Rural stretches: 30 minutes to 1 hour, depending on road access and distance from emergency facilities.

11. Overall Summary of Risk Assessment

This route involves mixed urban and rural environments, each presenting its own challenges. Traffic congestion, variable road conditions, and environmental considerations are key risks to monitor. Reliable contingency plans, adherence to local safety regulations, and awareness of communication dead zones are vital. Maintaining communication with local emergency services for timely assistance during incidents is recommended. Overall, proactive risk management and “defensive driving” principles should contribute to safe transport of hazardous materials along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.45358, 77.85781	30 KM/Hr
5	Turn	Medium	11.45353, 77.85697	30 KM/Hr
6	Turn	Medium	11.45839, 77.85143	30 KM/Hr
7	Turn	Medium	11.46053, 77.85034	30 KM/Hr
8	Turn	Medium	11.46307, 77.84941	30 KM/Hr
9	Turn	Medium	11.46316, 77.84921	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Blind Spot	Blind Spot	11.45631, 77.81668	10 KM/Hr
12	Blind Spot	Blind Spot	11.62145, 78.10740	10 KM/Hr
13	Turn	High	11.62130, 78.10776	15 KM/Hr
14	Blind Spot	Blind Spot	11.61217, 78.10439	10 KM/Hr
15	Turn	Medium	11.64844, 78.14338	30 KM/Hr
16	Turn	Medium	11.65152, 78.14229	30 KM/Hr
17	Blind Spot	Blind Spot	11.65573, 78.13972	10 KM/Hr
18	Turn	Medium	11.65678, 78.14605	30 KM/Hr
19	Turn	High	11.66305, 78.14758	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
14	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
17	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
18	clinic	Salem Care Foundation	11.6371946, 78.1313713	30 km/h	Medium
19	hospital	Aravind Eye Hospital	11.641927, 78.1392347	30 km/h	Medium
20	police	Annadanapatti B 4 Police Station	11.6424031, 78.1404779	30 km/h	Medium
21	clinic	Shri Bharthi Nursing Home	11.6420069, 78.1424099	30 km/h	Medium
22	hospital	Pramilagurunathan Hospital	11.6544063, 78.1464484	30 km/h	Medium
23	hospital	Shree Varma Ayurveda Hospital	11.656905, 78.14717	30 km/h	Medium
24	hospital	Nalam Hospital	11.65862, 78.147448	30 km/h	Medium
25	hospital	Sukumar Hospital	11.6601745, 78.1447516	30 km/h	Medium
26	clinic	Kathirveel Orthopeadic Hospital	11.6621451, 78.1473041	30 km/h	Medium
29	hospital	St.Marys Hospital, Salem	11.6634501, 78.1444402	30 km/h	Medium
30	clinic	Abhi Eye Hospital	11.6641165, 78.1457788	30 km/h	Medium
31	hospital	Nithish Ge And Multi Speciality Hospital	11.6636554, 78.1457305	30 km/h	Medium
32	clinic	Dr. S. S. Meera Vasudevan Clinic	11.6634071, 78.1457976	30 km/h	Medium
33	clinic	Sri Dhanvanthari Siddha Clinic	11.6634617, 78.1477869	30 km/h	Medium
34	hospital	Satya Sai Ent Head and Neck Hospital	11.664806, 78.1466813	30 km/h	Medium
35	hospital	Krishna Fertility and Laparoscopy Hospital	11.6643332, 78.1459989	30 km/h	Medium
36	hospital	Konlampatty Enocall Hospital	11.6642816, 78.1459647	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
37	clinic	Dr. Rajenderan Child Clinic	11.6643448, 78.1483264	30 km/h	Medium
38	clinic	Omega Clinic	11.660922, 78.150292	30 km/h	Medium
39	hospital	Prn Critical Care Centre And Hospitals	11.666386, 78.1493931	30 km/h	Medium
40	clinic	Salem Poly Clinic	11.6663605, 78.144658	30 km/h	Medium
41	hospital	S.P.S. Hospital	11.6669469, 78.1445952	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
15	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
27	school	St. Mary's Hr. Sec. School	11.662587, 78.1449627	30 km/h	Medium
28	school	Little Flower School	11.662629, 78.1495633	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45358, 77.85781



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45839, 77.85143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46053, 77.85034



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46316, 77.84921



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62145, 78.10740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61217, 78.10439



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64844, 78.14338



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65152, 78.14229



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.65573, 78.13972



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65678, 78.14605



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66305, 78.14758

